Role: Electronics Hardware Engineer
Location: Munich, Germany / San Francisco, CA

Case Study Instructions

Design a Camera Module for Raspberry Pi Zero 2 W

Objective

Design a daughter board for a Raspberry Pi Zero 2 W that integrates the IMX219 camera module, a PIR motion sensor, and an ambient light sensor. The daughter board should match the size of the Raspberry Pi Zero W and be powered by a standard Raspberry Pi supply.

In the case study panel presentation, you'll have 45 minutes with the panel. Plan for about 15/20 minutes of presentation time, as the panel will ask questions during this session.

Please prepare a presentation for this panel that introduces the challenge in the prompt outlined below, along with your recommended plan to move forward. As you work through questions or new variables introduced by the panel, feel free to whiteboard live or even change your final recommendation while in the case study panel together.

You should start the presentation by introducing yourself to the audience and include relevant information about you and your professional experience (max: 2 minutes).

Task Outline

1. **System Requirements**
    - **Camera Module**: Integrate the IMX219 camera module (IQL-IMX219/FF).
    - **Motion Detection**: Integrate a suitable PIR motion sensor to detect motion to trigger the camera.
    - **Lighting Adjustment**: Integrate a suitable ambient light sensor that will be later used to adjust camera settings based on lighting conditions.
    - **Power Supply**: Design a power supply circuit to ensure stable operation. Choose and justify the power management components used. Use at least one switching regulator in your design.
    - **Form Factor**: Ensure the daughter board fits within the dimensions of the Raspberry Pi Zero 2 W (65mm x 30mm).
    - **Interconnection**: The daughter board will connect to the Raspberry Pi Zero 2 W through the GPIO header. All components, including the camera module, PIR sensor, and

ambient light sensor, must be placed on the daughter board. Additionally, include an FFC connector on the edge of the daughter board to allow the user to connect an FFC cable between the daughter board and the CSI interface of the Raspberry Pi Zero W.

- **Power Supply**: Assume the Raspberry Pi will be powered with a 5V 3A DC power supply. Consider power losses and efficiencies.

2. **Design Tasks**
    - **Schematic Design**: Create a comprehensive schematic diagram including all components.
        - **Camera Module (IMX219)**
        - **PIR Motion Sensor**
        - **Ambient Light Sensor**
        - **Power Supply Circuit**
        - **Connectors for Raspberry Pi GPIO and CSI interface**
    - **PCB Layout**: Design a PCB layout that matches the dimensions of the Raspberry Pi Zero 2 W.
        - Ensure minimal noise interference between analog and digital components.
        - Provide efficient power routing to reduce energy losses and ensure stable operation.
    - **Power Management**: Select and justify appropriate voltage regulators to provide stable power to all components.
    - **Design for Testability**: Implement features that facilitate the testing and validation of the board during and after manufacturing.
    - **Design Documentation**: Provide a brief but comprehensive documentation of your design decisions.
        - Explain the choice of the PIR motion sensor and its placement.
        - Justify the selection of the ambient light sensor and its integration.
        - Detail the power supply design, including regulator choices and their specifications.
        - Describe considerations for minimizing noise and ensuring reliable operation.

3. **Deliverables**
    - **Schematic Design**: Complete the schematic diagram in PDF or a design tool file (e.g., Eagle, Altium, KiCAD).
    - **PCB Layout**: Design files, Gerber files, and a 3D render of the PCB as a Complete design package
    - **Component List**: Bill of materials (BOM) with detailed component specifications.
    - **Power Supply Design**: Detailed documentation of the power supply design, including regulator choices and justifications.
    - **Design Decisions Documentation**: Comprehensive explanation of design choices and considerations.